

Modular Normlight System Based on Spectral Reconstruction and Sensor-Based Verification

Abstract

This white paper describes a modular normlight and illumination system intended for professional visual color assessment in contexts where spectral accuracy, photometric adequacy, and long-term reproducibility must be demonstrable, traceable, and maintainable over time. Rather than approximating illuminant conditions through a single broadband source defined primarily by correlated color temperature, the system generates lighting conditions through spectral reconstruction. In this approach, target illuminants are synthesised from multiple independently controllable spectral channels whose combined spectral power distribution is actively shaped to match defined reference conditions.

The system architecture is based on autonomous light modules that can function as independently calibrated light sources or be combined into larger configurations, such as normlight cabinets or spatial luminaires. This modularity allows luminous output to be scaled to application geometry, surface dimensions, and viewing distance, while preserving calibrated spectral composition and avoiding degradation of spectral integrity at higher output levels.

Scientific credibility and professional applicability are ensured through bandwidth-aware channel design, factory calibration, sensor-based verification, and boundary-driven recalibration. In this framework, recalibration is triggered only when measured deviations exceed predefined tolerance limits, rather than through continuous adjustment or fixed periodic maintenance schedules. This approach balances long-term stability with operational robustness.

Beyond single-unit operation, the system explicitly supports cross-unit calibration and fleet harmonisation. Multiple installations can be aligned to a shared spectral reference, enabling distributed visual assessment environments to maintain consistent illumination conditions across locations without requiring identical hardware configurations or centralized operational control.

Networked Calibration and Fleet Harmonisation Architecture

The system is designed to support both standalone deployment and coordinated operation across multiple installations. Individual modules or complete assemblies may operate independently, while still participating in a broader calibration framework that maintains spectral and photometric consistency across a distributed fleet.

This architecture addresses use cases in which visual color assessment must remain reproducible not only over time at a single location, but also across multiple sites, such as

geographically distributed quality control facilities, multi-site production environments, or supply-chain assessment contexts where visual decisions must be comparable across organizations.

Within a networked configuration, one unit is designated as a reference unit, while other units operate as fleet members. The reference unit establishes the normative spectral and photometric targets to which fleet members align their calibration. Importantly, this reference–member relationship is calibrational rather than operational in nature. Fleet members remain fully autonomous during normal lighting operation, and no real-time control signals are exchanged between units during use.

Synchronization is limited to calibration parameters, spectral targets, tolerance definitions, and lifecycle metadata. As a result, network outages or connectivity limitations do not impair lighting functionality, and individual units continue to operate safely and predictably even in complete isolation. At the same time, calibration updates can be propagated systematically and auditable alignment across installations can be maintained.

Fleet-level calibration operates at two complementary levels. Each unit undergoes factory calibration and periodic local verification to establish and maintain its own baseline spectral and photometric characteristics. In addition, relative calibration is performed at the fleet level, whereby fleet members verify their output against the reference unit's published targets. When deviations exceed predefined tolerance boundaries, recalibration is triggered to restore alignment with the fleet reference.

The system architecture supports both composite spectral power distribution matching and channel-resolved calibration. In the former approach, the combined spectral output is aligned to a reference distribution across the visible spectrum. In the latter, individual spectral channels are calibrated independently, allowing component-level alignment while preserving modular channel control. The architecture accommodates both strategies without prescribing a single method, allowing implementation choices to be adapted to application requirements.

Data exchange within the fleet is intentionally constrained. Shared information may include spectral target definitions, colorimetric tolerance boundaries, photometric targets, calibration timestamps, validity periods, and firmware version identifiers. Real-time lighting states, operational commands, continuous sensor data streams, and user interface settings are explicitly excluded from fleet exchange. Fleet coordination thus remains focused on calibration alignment and long-term consistency management, while operational autonomy is preserved at all times.

Connectivity Architecture

Network connectivity is designed to enhance calibration synchronisation, firmware management, and lifecycle coordination without introducing operational dependencies. The system remains fully functional as a lighting solution in the absence of network access.

Wired Ethernet serves as the primary communication layer, providing deterministic and robust connectivity suitable for professional and industrial environments. In addition, wireless connectivity is supported through Wi-Fi operation in two modes. In access-point mode, the unit provides a local wireless network for configuration and service access without reliance on external infrastructure. In client mode, the unit integrates into an existing wireless network for fleet coordination and lifecycle management.

Bluetooth connectivity may be provided for configuration or diagnostic purposes, but it is not architecturally required and does not form part of the core system functionality.

Loss of network connectivity does not impair lighting output, spectral characteristics, sensor-based verification, preset recall, or boundary-driven recalibration logic. Connectivity enhances fleet-level consistency and lifecycle efficiency, but individual units remain fully autonomous by design.

Firmware and Lifecycle Management

System firmware governs spectral channel control, calibration algorithms, sensor integration, and communication protocols. Firmware is maintained centrally and distributed to units via network connectivity, supporting coordinated evolution of calibration methodology across fleets.

Firmware updates are issued infrequently, typically one to two times per year, and address calibration algorithm refinements, sensor integration improvements, communication protocol updates, and user interface evolution. Updates are performed remotely without physical access to the unit. Integrity verification is performed prior to installation, rollback capability is maintained in case of failure, and update status is logged for traceability.

In fleet configurations, firmware versioning supports staged deployment. Reference units may be updated first, followed by controlled rollout to fleet members, ensuring that calibration targets remain interpretable and consistent throughout transitions.

IP Positioning and Differentiation

The novelty of the system resides in the integration of autonomous modular architecture with independent thermal and spectral control, sensor-based verification as an intrinsic system function, bandwidth-aware channel design with explicit consideration of spectral overlap, boundary-driven recalibration, and networked fleet architecture enabling inter-unit calibration harmonisation without centralized operational control.

In contrast to conventional normlight solutions, which are calibrated exclusively as isolated installations, this system explicitly supports calibration alignment across distributed environments. This capability addresses scenarios in which visual consistency across

locations is operationally critical, such as distributed quality control, multi-site production coordination, and supply-chain assessment.

This document establishes the system principle as prior art and conceptual delineation without disclosing proprietary channel compositions, specific algorithms, communication protocols, or implementation-specific tolerance schemes.

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